DLD (SY EXTC)(Sem III)
July - Oct. 2025
ISE paper solution & marking scheme

Q.1 a) 1) $(7)_8 - (4)_8$ using 2's complement method.

$(7)_8 = (111)_2$ $\quad$ 111 $\quad$ 1 m

$(4)_8 = (100)_2$ $\quad$ − 100

1's complement of 100 is 011

2's complement of 100 is $\begin{array}{r} 011 \\ +\ \ 1 \\ \hline 100 \end{array}$ $\quad$ 1 1/2 m

$\therefore$ $\begin{array}{r} 111 \\ +\ 100 \\ \hline 1\,011 \end{array}$

end around carry

$\boxed{\text{Answer is } (011)_2}$

$\boxed{(011)_2 = (3)_8}$ $\quad$ 1 m.

2) Convert $(2A5)_H$ into decimal, octal & BCD

$(2A5)_H = (677)_{10}$ (1 m)

$(2A5)_H = \underline{001}\,\underline{010}\,\underline{100}\,\underline{101}$

$(2A5)_H = (1245)_8$ (1m)

$(2A5)_H =$

2 A 5

$2\times(16)^2 + 10\times(16)^1 + 5\times(16)^0$

$= 2\times256 + 160 + 5$

$= 512 + 165$

$= 677$

$(2A5)_H = (677)_{10}$ (1 1/2 m)

$6 = 0110$

$7 = 0111$

$7 = 0111$

$\therefore$ BCD of $(2A5)_H$ is 011001110111 (1 1/2 m)

3) Gray 10110 to binary code

$g_4\, g_3\, g_2\, g_1\, g_0$

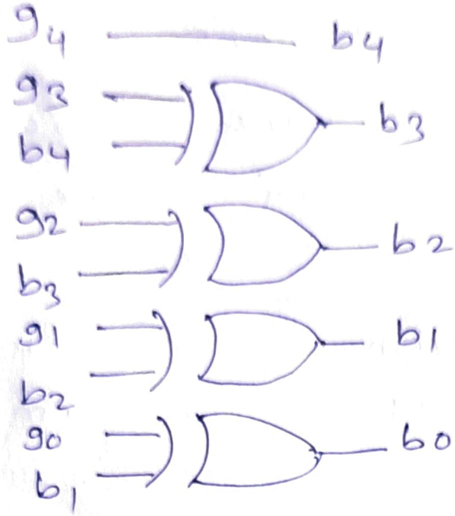
1 0 1 1 0

1 1 0 1 1

$b_4\, b_3\, b_2\, b_1\, b_0$

Binary code of 10110 Gray code is $\boxed{11011}$

ckt implementation dia.



(1m)

4) BCD addition of $(9)_{10} + (5)_{10}$

BCD of $(9)_{10} \rightarrow 1001$
BCD of $(5)_{10} \rightarrow 0101$ (1m) $(9)_{10} + (5)_{10} = (14)_{10}$

$$\begin{array}{r} 1001 \\ +\ 0101 \\ \hline 1110 \\ +\ 0110 \\ \hline 10100 \end{array}$$

1110 ← Invalid BCD — (1m)

10100 ← Valid BCD. (1/2 M)

Answer is (00010100) BCD

1 (b) Diagram of 4 bit controlled binary adder / subtractor — (5m)

Q. 2 (a) 1) $F(a,b,c) = \Pi M(0,2,4,6,7)$

a \ bc	00	01	11	10
0	0			0
1	0		0	0

(1 1/2 m) $= c \cdot (\bar{a} + \bar{b})$ (1m)

2) $F(p,q,r) = \Sigma m(0,1,3,5)$

p \ qr	00	01	11	10
0	1	1	1	
1		1		

$= \bar{p} \cdot \bar{q} + \bar{p} r + \bar{q} r$ (1m)

(1 1/2 m)

$Y = \Sigma m(1,3,7,11,15) + d(0,2,5)$

AB \ CD	00	01	11	10
00	X (0)	1 (1)	1 (3)	X (2)
01	(4)	X (5)	1 (7)	(6)
11	(12)	(13)	1 (15)	(14)
10	(8)	(9)	1 (11)	(10)

Solution 1

$2\frac{1}{2}$m $Y = \bar{A}D + CD$ - $2\frac{1}{2}$M

OR

AB \ CD	00	01	11	10
00	1	1	1	1
01		1	1	
11			1	
10			1	

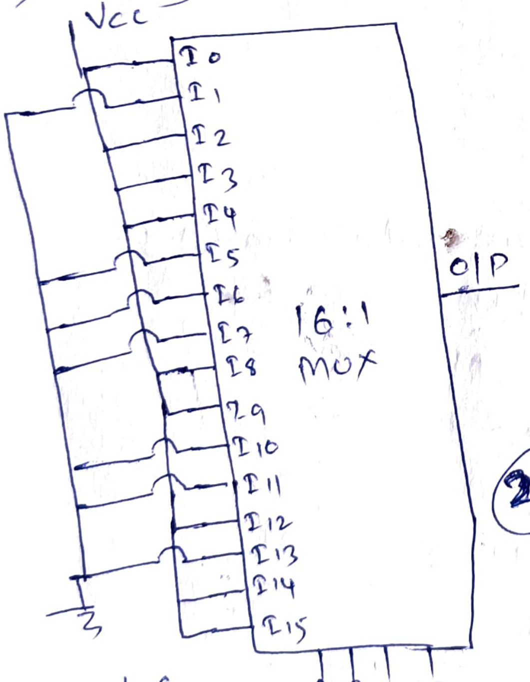
$2\frac{1}{2}$m

Solution 2

$Y = \bar{A}\bar{B} + \bar{A}D + CD$ $2\frac{1}{2}$m

Q.2(b) $Y = F(A,B,C,D) = \Sigma m(0,2,3,4,8,9,12,14,15)$

1) Using single 16:1 Mux



2M

2) Using single 8:1 mux

I/PS				O/P
A	B	C	D	Y
0	0	0	0	1
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1
1	0	1	0	0
1	0	1	1	0
1	1	0	0	1
1	1	0	1	0
1	1	1	0	1
1	1	1	1	1

2 m

$I_0 = \bar{D}$

$I_1 = 1$

$I_2 = \bar{D}$

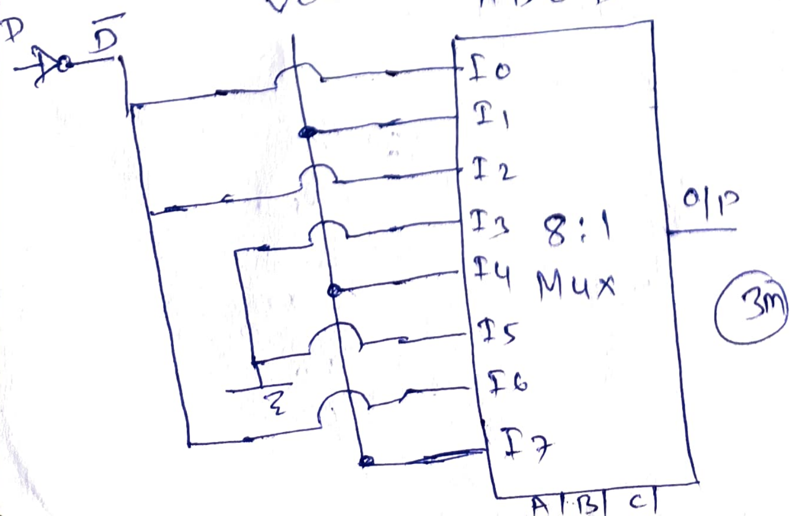
$I_3 = 0$

$I_4 = 1$

$I_5 = 0$

$I_6 = \bar{D}$

$I_7 = 1$

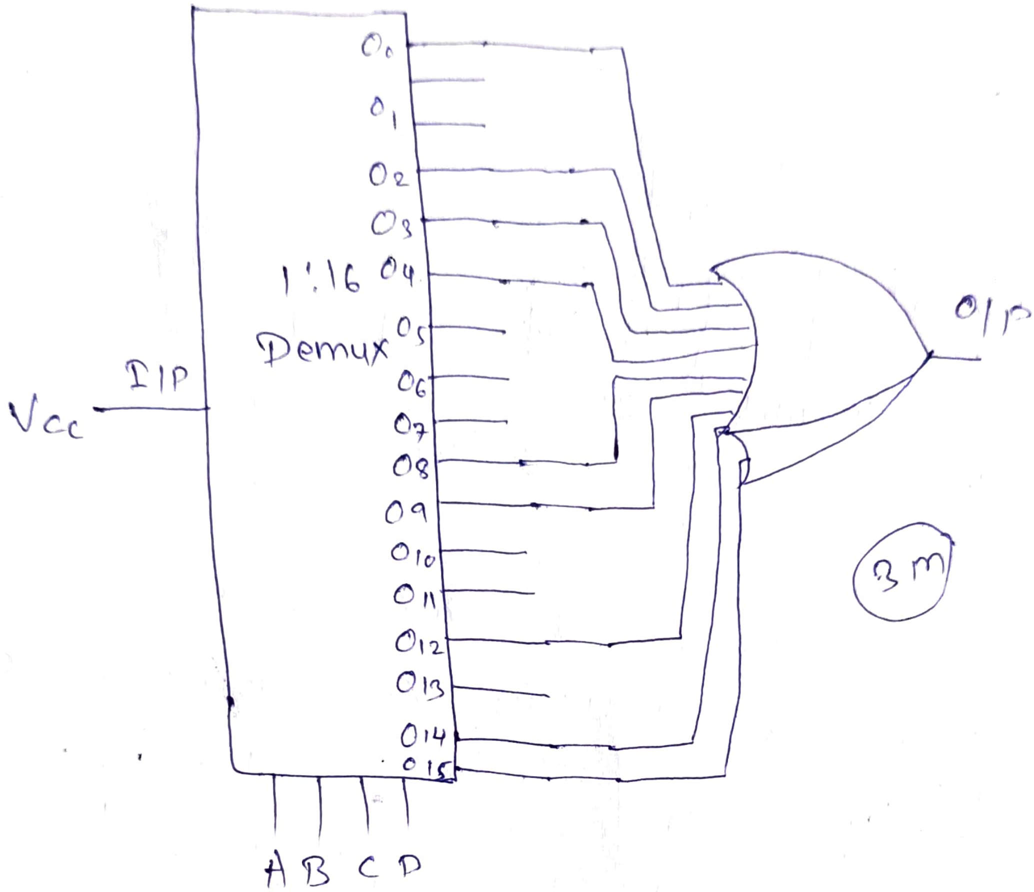


3m

Do not write/Mark on Bar

(4)

(3) Using 1:16 Demux



(3M)

Q.3

SR FF using NOR gate dia - 1 M
Truth table - 1 m
Working - 3 M

OR

Diagram of JK FF with - 1 m
-ve edge clock
Truth table - 1 m
Working - 3 M.